

Week 5 Quiz

Multiple Choice

1. [5 points] Consider the following function:

$$f(x) = x \ln(x) + e^x x^2.$$

What is the value of the derivative $f'(x)$ at the point $x = 0$?

- (a) $f'(0) = -\infty$
 - (b) $f'(0) = 0$
 - (c) $f'(0) = -0.5$
 - (d) $f'(0) = -1$
2. [5 points] What is the 2026'th derivative of the function $\cos(-2x)$?
- (a) $(-2)^{2026} \sin(x)$
 - (b) $-2 \cdot (-1)^{2026} \sin(-2x)$
 - (c) $-(2026)^{-2} \cos(-2x)$
 - (d) $-(2)^{2026} \cos(-2x)$

Written Response

3. [10 points] Differentiate the following function in the variable x . Do not simplify your answer.

$$f(x) = \frac{\cos(x) + 1}{\cos(x)}$$

Answer Key

1. **Answer:** (a.)

Answer: (d.)

Answer:

$$\frac{-\cos(x)\sin(x) + (\cos(x) + 1)\sin(x)}{\cos^2(x)}$$

Rubric for 3: 0 points for very wrong answer.

2 points for the answer 1

6 points if showed knowledge of quotient rule but incorrect.

10 points for correct answer.